Technology that Extends the Service Life of Concrete Structures

MCI®

MIGRATING CORROSION INHIBITORS
FROM GREY TO GREEN



CORTEC CORPORATION
Environmentally Safe VpCI®/MCI® Technologies

ISO/IEC-17025 LABORATORY TESTING CERTIFIED

ENVIRONMENTAL ISO 14001 CERTIFIED SYSTEM REGISTERED

ISO 9001 CERTIFIED QUALITY SYSTEM REGISTERED

A Revolutionary Way to Extend the Service Life of Concrete Structures. Simple...Sure...Safe.

Cortec's patented MCI® (Migrating Corrosion Inhibitors) technology protects reinforcing metal in concrete from corrosion. MCIs rehabilitate existing concrete structures and extend the life span of new structures. Often, corroding rebar in deteriorating concrete is the cause of costly repairs, financial losses, injuries and deaths, but Cortec® has the corrosion solution. Cortec® MCI® products for concrete maintain structural integrity, rehabilitate vulnerable structures, and alleviate environmental concerns. A unique feature of MCI® is that the inhibitor will migrate a considerable distance through concrete to protect embedded ferrous metals.

Causes of Corrosion:

Reinforcement in new concrete is generally protected from corrosion due to the high alkaline nature of the concrete itself. The high pH of the concrete (usually greater than 12.5) causes a passive oxide film to form on the steel. Environmental factors can affect this protective oxide film and induce the formation of corrosion cells. Once corrosion starts, some parts of the reinforcement become anodic, discharging iron ions (current) into the electric cell. Steel areas that receive this current are the cathodic areas of the corrosion cell. This is where hydroxide ions are formed. Iron and hydroxide ions react to form iron hydroxide, FeOH, which further oxidizes to form rust. Once started, the rate of corrosion is affected by the concrete's electrical resistivity, moisture content, and the rate at which oxygen migrates through the concrete to the steel. As rust formation continues, it can take up to four times the volume originally occupied by the embedded reinforcement, causing cracking and spalling of the concrete.

Chlorides:

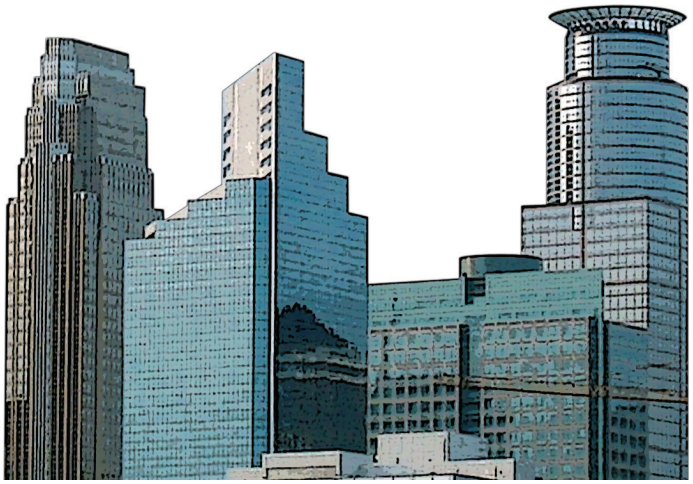
Chloride ions can penetrate the passive oxide film on the reinforcement. They combine with iron ions to form a soluble iron chloride complex that carries the iron into the concrete for later oxidation (rust). Once chlorides reach a level of about 0.15% (water soluble chloride by mass of cement) in the concrete, corrosion starts. Concrete can be exposed to chlorides from several different sources, including chloride containing set accelerators, deicing salts, seawater, and airborne salts.

Carbonation:

Carbonation is the process by which carbon dioxide in the air reacts with hydroxides in the concrete such as calcium hydroxide, to form carbonates. This reaction significantly lowers the pH. When the pH of concrete surrounding embedded reinforcing steel drops below 12, the protective oxide layer is lost, and the corrosion process begins.

Acid Rain/Industrial Pollutants:

Acids attack concrete by dissolving the cement paste and calcareous aggregates. They also reduce the pH of the concrete, allowing the corrosion process to begin. Pollutants such as sulfate attack the concrete by reacting with hydrated compounds in the hardened cement paste. These reactions can lead to disintegration of the concrete, making embedded reinforcement more susceptible to corrosive attack.



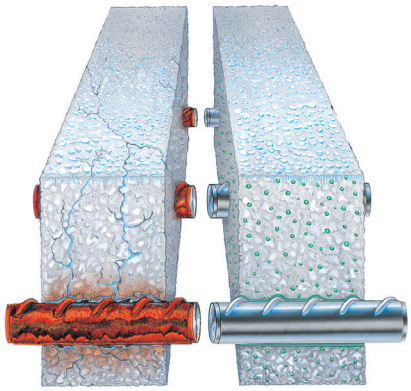
Once a concrete structure is built, it's impossible to coat the reinforcing steel with fusion-bonded epoxy to protect it from corrosion. Cathodic protection is ineffective unless the steel reinforcement is electrically continuous.

Cortec® MCI®, however, can be easily added to new concrete or used for rehabilitation and will not delay construction or increase construction costs other than the small cost of the material. Unlike standard inorganic inhibitors, Cortec® MCIs do not have to come in contact with the reinforcing steel upon application because they can migrate to the steel and protect it.

When specified in new construction, Cortec's MCI® line of concrete admixtures offers reinforcing steel superior corrosion protection against carbonation and chloride attack.

Comparison of Cortec® MCI® Admixtures to Other Inhibitors:

Feature	Cortec® MCI® Inhibitor	Calcium Nitrite:
Environmentally friendly, derived from renewable resources	True	False
Used in small quantities—less than 1.5 pints/yd3 (1 liter/m3)	True	False
Required dosage rate is not affected by expected chloride exposure	True	False
Ability to migrate through concrete in a vapor phases at ambient temperatures	True	False
Does not increase shrinkage compared to a control	True	False
Does not require adjustments to concrete mix design (chemical or water)	True	False
Does not affect concrete resistivity	True	False
Does not accelerate concrete set time	True	False
Has UL approval to meet NSF Standard 61 (contact w/potable water)	True	False
Spills can be flushed with large quantities of water down drain	True	False





Summary of Data Involving Cortec® MCI® 2005 NS in Concrete

	Control	MCI® 2005 NS
Air Content (AEC)*	6.2%	6.3%
(NEAC)	2.7%	2.5%
Std. Dev.	1.8	2.4
Compressive Strength (psi) 6 x 12 inch cylinders		
@7 days (AEC)	3550	3630
(NAEC)	3610	4180
@28 days (AEC)	4760	5030
(NAEC)	5180	6120
Flexural Strength, psi, ASTMC192, C78		
@7 days (AEC)	610	610
(NAEC)	720	720
@28 days (AEC)	720	720
(NAEC)	820	820
Setting Time, ASTM C403		
Setting Time, Hour: Minutes		
Initial (AEC)	3:28	3:37
(NEAC)	3:35	3:35
Final (AEC)	4:55	5:07
(NEAC)	4:58	4:55
Slump (AEC)	3 1/3”	3 1/4”
(NEAC)	3 1/4”	3”
Freeze Thaw, ASTM C666, Method A		100%
Relative Durability		

*AEC = air entrained concrete
NAEC = non-air entrained concrete

How Does MCI® Technology Work?

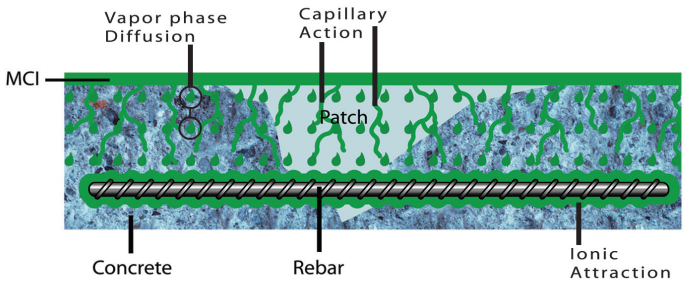
The corrosive effects of carbonation and chlorides cause a breakdown of the natural passivating layer on steel in concrete. MCI's provide protection because of their ability to migrate to the depth of the metal, and form a protective, molecular layer on steel when they come into contact with it.

1. MCI® move as a liquid into the concrete matrix. In new construction, MCI® is admixed either with the batch water or directly into a mixer. For existing structures, MCI applied to the surface is drawn into the concrete via capillary action—the concrete acts like a sponge, drawing MCI inside.

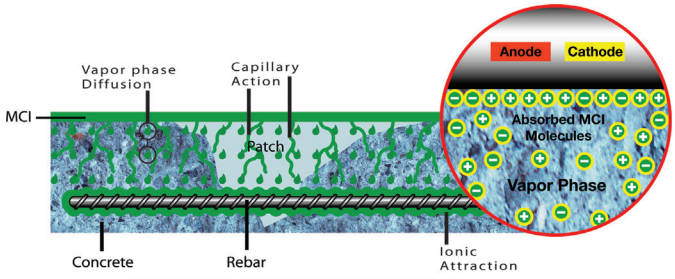
2. MCI® move in a vapor phase throughout the concrete pore structure. This movement is governed by Fick's Law, meaning molecules move randomly throughout the matrix from areas of high concentration to areas of low concentration.

3. When MCI® come into contact with steel, it has an ionic attraction to it, and forms it's protective, molecular layer. MCI's affinity to the metal is stronger than water, chlorides and other corrosive contaminants.

4. Independent testing has confirmed that MCI can adsorb onto metal to a depth of 75-85 nm, forming a layer that is between 20 and 100 Å thick. In the same testing, chlorides were shown to penetrate only 60 nm deep. This confirmed the ability of MCI to displace chlorides on the metal surface and provide protection even in their presence.



MIGRATING CORROSION INHIBITORS



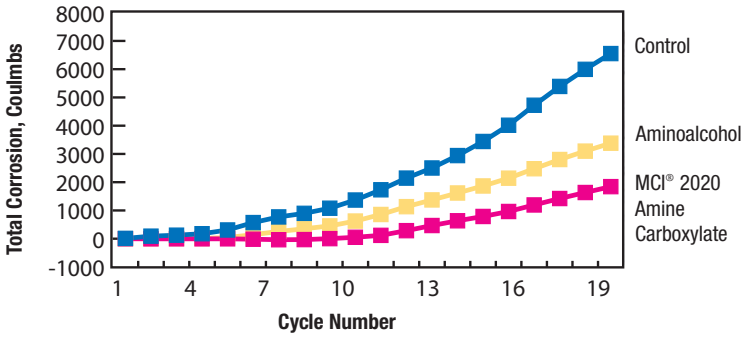
MIGRATING CORROSION INHIBITORS

Competitor Comparisons

Time and time again, MCI® products are shown to outperform the competition. Whether using surface applied MCI® 2020 on existing structures to extend the life of a repair or using MCI® 2005 series admixtures to greatly increase the expected service life of a new structure, you can be sure that Cortec® MCI® will provide you with superior corrosion protection.

Cracked Beam Testing of Surface Treatments

Amine Based Products
(3.5% NaCl)

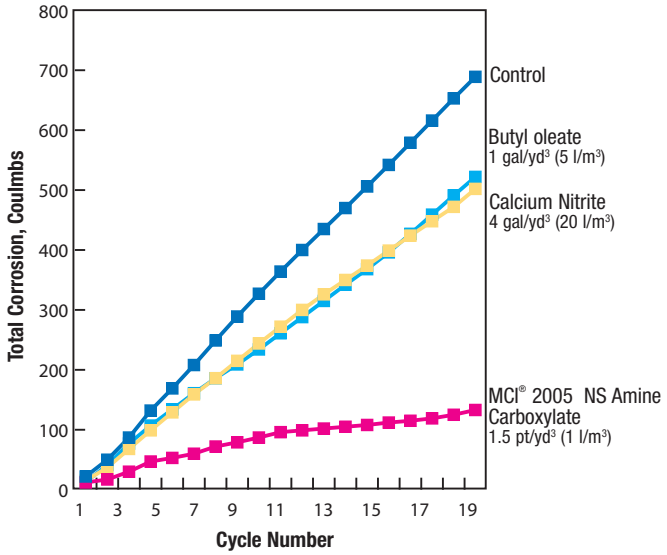


Above: Wiss, Janney, Elstner Associates, Inc.
January 1995, WJE No: 922041

Right: American Engineering and Testing
August 2003, AET Job No:05-01171

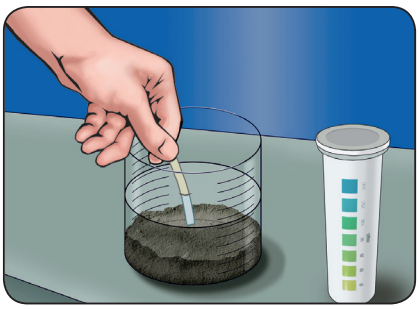
Cracked Beam Testing of Admixtures

MCI® 2005 NS vs. Competition
(6% NaCl)

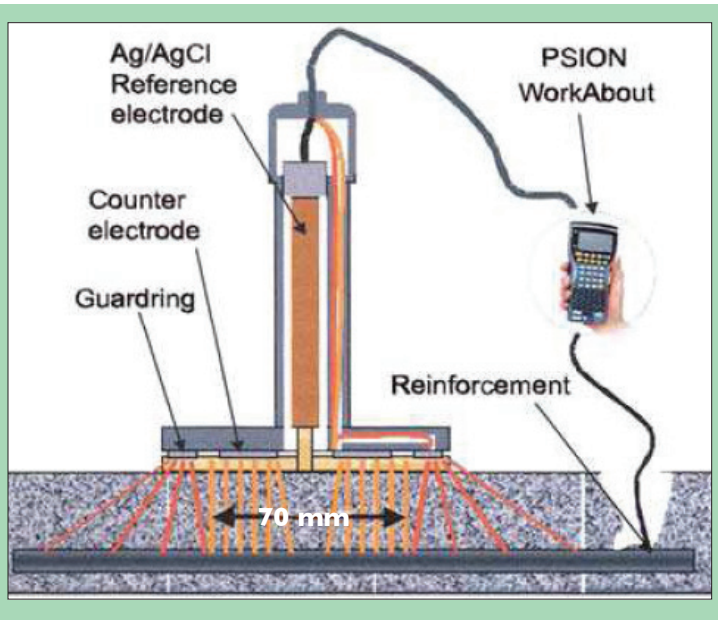


Testing of Migrating Corrosion Inhibitors

The effectiveness of Migrating Corrosion Inhibitors can be monitored in many ways. A Quaternary Ammonium Compounds (QAC) test kit is a simple test method to show the presence of our surface applied MCI® in concrete structures. QAC's are a component of the MCI® product chemistry. While this test is not quantitative, it can verify the presence of inhibitor in the structure and can be monitored to show migration of MCI® to the depth of the reinforcement.

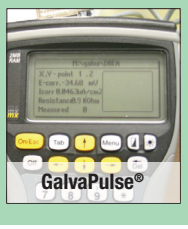


Use the manufacturer's instructions for the EM Quant QAC test sticks to analyze each slurry solution/extraction.



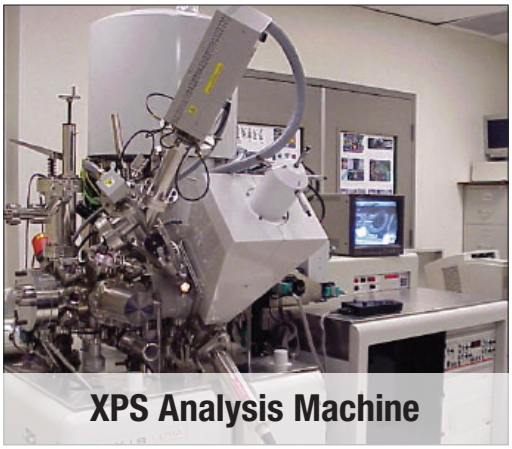
Testing the effectiveness of MCI® can be done by monitoring actual corrosion rates of embedded reinforcement. A control reading can be taken, and then subsequent readings can be performed after application of MCI®. This will show what the rate was initially, how low it drops after application of inhibitor, and allow you to determine when you should reapply more inhibitor for additional corrosion protection. There are several different pieces of equipment that use linear polarization to monitor corrosion rates of metallic reinforcement in concrete. These include: Gecor® 6, Gecor® 8, GalvaPulse®, and embedded corrosion rate sensors such as Corrator Probes.


Corrator Probes

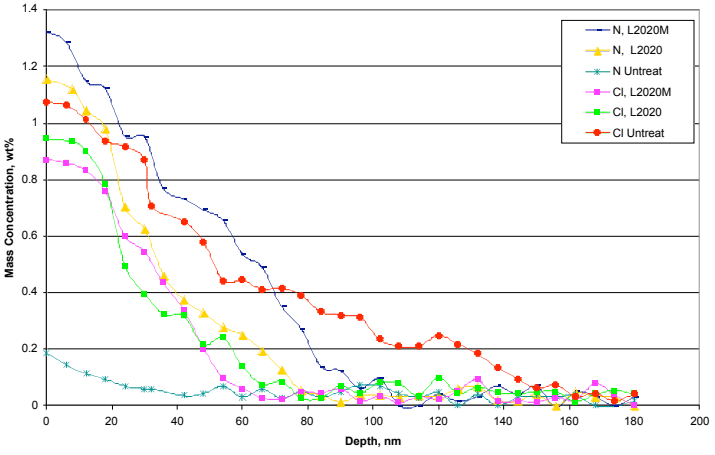

GalvaPulse®


Gecor® 6

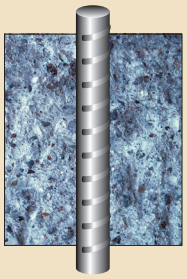
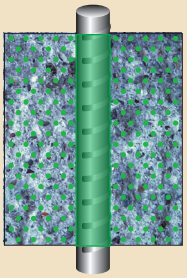
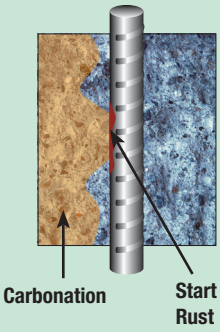
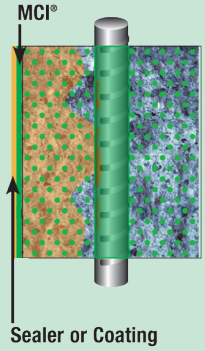
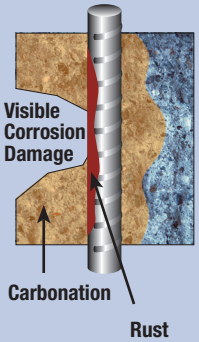
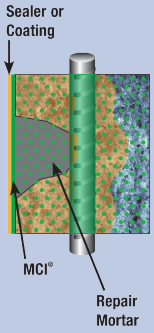
The molecular bond MCI® forms on embedded metals has been verified using X-ray Photoelectron Spectroscopy (XPS) which takes a picture of the metal and can show the depth at which molecules adsorb onto it. This test showed that not only did the MCI® migrate through the concrete, it also adsorbed onto the metal to a deeper depth than chlorides, effectively mitigating corrosion. Contact Cortec® Technical Service if you need more detailed information on testing Migrating Corrosion Inhibitors.


XPS Analysis Machine

XPS Depth Profile (Ar at 4 kV, 15 mA)
Untreated, Inhibitor A and B Concrete Samples after 450 days of testing



MCI® Product Application Guide

Condition of Structure	Objectives & Requirements	MCI® Protection	Features & Benefits
STAGE 1 New Concrete			
 • Aggressive Environment • Insufficient Concrete Cover	• Extend useful service life • Protect from premature corrosion • Preserve the natural appearance of the concrete	 • MCI® 2005 series admixtures can double to triple the time to corrosion initiation, and once corrosion starts, they can cut rates by more than 5 times compared to a control	• Low Dosage Rate • UL Certified to meet NSF Standard 61 Requirements • No affect on concrete mix design • No affect on concrete properties • Can double the service life of many new structures
STAGE II Existing Structures, No Visible Corrosion Damage			
 • Concrete structures without protective coatings • Aggressive environment • Initiation of corrosion • No spalling or cracking	• Slow the rate of corrosion • Protect against possible concrete damage • Protect against further corrosion due to carbonation and/or chloride penetration	 • Application of a MCI® 2020 Series surface applied product by spray, brush or roller • Followed by an application of an anticarbonation coating such as MCI® Architectural Coating OR application of a sealer such as MCI® 2018, 2019, 2021, or 2022	• High coverage rate • Minimal or no concrete removal • Non-destructive • Extends the time to next repair of the structure • Fewer coats means lower labor costs than competitor products • Can be 10 times less costly than a Stage III repair! • MCI® 2020 Series meets NSF Standard 61 requirements
STAGE III Existing Structures, Visible Corrosion Damage			
 • Concrete surface with visible corrosion damage (i.e. spalling and cracking), repairs are necessary • High level of chlorides at depth of reinforcement	• Repair of damaged surfaces • Long term protection against future exposure of contaminants • Enhanced protection against the continuing damage of latent corrosion • Reduced risk of ring-anode (insipient anode) effect	 • Cleaning of exposed reinforcement with Cortec's VpCI® 423, or use of Cortec's VpCI® CorrVerter® • Application of Cortec's MCI® 2023 grout to exposed reinforcement and repair area • Application of Cortec® MCI® 2039 repair mortar • Application of Cortec® MCI® 2038 Finish repair mortar • Application of Cortec's MCI® 2020 to entire surface area • Application of Cortec® Coating or Sealer	• Aesthetically pleasing restoration of structure to a safe condition • Complete repair and protection against latent corrosion damage • Can more than double the life of the repair (based on G109 testing) • MCI® 2020 Series is UL certified to meet NSF Standard 61 requirements

Relevant Case Histories

Construction of Wells Fargo Parking Garage, MCI® 2005 NS (214)



An 1,800 vehicle, six-level parking garage needed all of it's 22,000 cubic yards of concrete to meet or exceed 3,000 psi strength within 18 to 24 hours.

A calcium nitrite based corrosion inhibitor didn't meet the required 24 hour minimum strengths, and also had shrinkage cracking. MCI® 2005 NS met the required specifications, reduced shrinkage by 30%, and eliminated shrinkage cracking even in sub-zero temperatures.

See also:
- China Railroad Bridge, MCI® 2000, 2020, 2021 (092)
- Concrete Bridge Foundation, MCI® 2000, 2020 (119)
- Concrete Railway Bridge, MCI® 2005, 2020, 2023 (139)
- City Street Bridge Deck, MCI® 2000 (211)
- Construction of Parkway Technology Campus, MCI® 2007 Super Corr™, 2022 Sealer (238)

Pentagon: Restoration of All Exterior Walls MCI® 2020 V/O (046)



Corrosion of embedded reinforcing steel was causing spalling on the walls. Carbonation (up to 3.5 in) on the walls lowered the pH of the concrete causing the corrosion.

The requirements included: a minimum 20 year design life, stop water absorption, reduce or stop corrosion, and maintain the appearance of the walls. The repair program consisted of 200,000 ft^2 of surface hand patch repair and over 1,000,000 ft^2 treated with MCI® 2020 V/O, and a silicate based coating.

MCI® 2020 V/O was chosen to protect and repair the walls based on its warranty and its fulfillment of the other specified repair design requirements.

See also:
- China Railroad Bridge, MCI® 2000, 2020, 2021 (092)
- Parking Ramp, MCI® 2020 (121)
- Concrete Railway Bridge, MCI® 2005, 2020, 2023 (139)
- Construction of the World's Largest Mosque, MCI® 2021, MCI® Architectural Coating White (236)
- Inland Steel Building, MCI® 2020 (263)

DePere Waste Water Treatment Tanks MCI® 2020, 2023, 2038 (219)



The outdoor waste water tanks were originally constructed in 1939, and no repairs had been made in almost 20 years. Corroding and spalled areas, as well as exposed rebar had to be repaired.

Concrete was sandblasted and pressure washed. Exposed rebar was coated with MCI® 2023 and spalled areas were repaired with MCI® 2038. This was followed by an application of MCI® 2020 to the entire surface area of the tanks.

Customer was very satisfied with the application and products used. They purchased more products to repair other tanks after the success of this project.

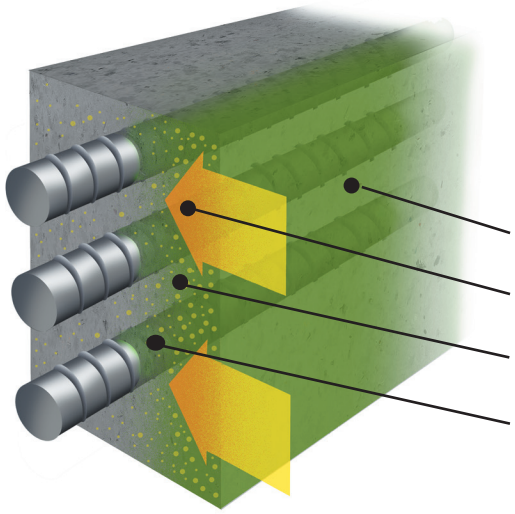
See also:
- HPRS™: High Performance Repair Systems, MCI® 2020, 2023, 2038, 2039, VpCI-386 (150)
- Concrete Spalling Due to Hydrochloric Acid Fumes, MCI® 2020, 2038, CorrVerter® (217)
- Leaking Parking Garage, MCI® 2020, 2022, 2023, 2038 (218)

Diffusion Through Concrete

Migrating Corrosion Inhibitors for Reinforced Concrete

B. A. Miksic, FNACE, D. Bjegovic, L . Sipos
(printed in ConChem-Journal, 2/93)

The diffusion rate of MCI® for two different types of concrete was determined using the mix designs (Table 1). Two concrete specimens were placed into a specially designed diffusion cell where they acted as a membrane between two solutions for a 21-day test period (Figure 1). For optimal results, the concentration MCI® was tested at three-day intervals. Every three days, a 15mL aliquot was taken for concentration determination, and after each measurement, the fluid was returned to the cell. An ORION 95-12 Ag/AgCl electrode containing 0.1 M NH_4Cl solution was used for determination of MCI® concentration. Concentration of the inhibitor in the concrete over time was found using electrode potential readouts and a calibration diagram (Figure 2).



MCI® concentration increases with time, showing that the MCI® diffuses through the concrete (Figure 2). Using experimentally obtained data, the coefficient of diffusion was calculated (Figure 3 and Table 2). This data demonstrates the migratory nature of MCIs and proves that they can be used for protection against chloride induced corrosion and carbonation.

General Composition		T-1	T-2
Components	Units		
Concrete	kg/m³	380	380
Water	l/m³	209	171
Aggregate	kg/m³	1720	1823
W/C	%	0.55	0.45
Consistency of Settling	cm	14.5	4.5

Table 1 Composition of Concrete

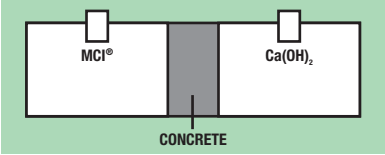


Figure 1 Diffusion Cell

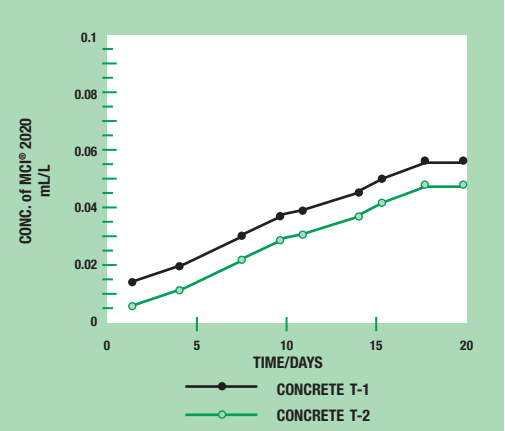


Figure 2 Calibration Diagram

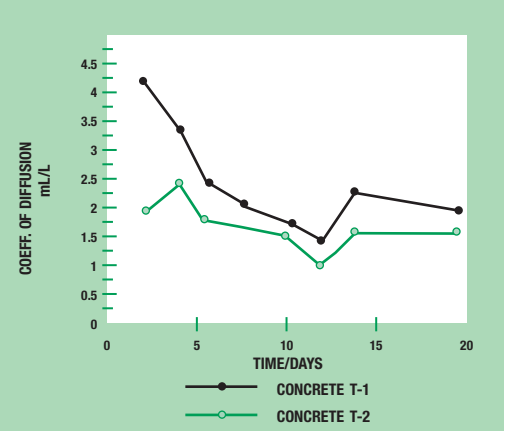


Figure 3 Calibration Diagram

Coefficient of Diffusion cm^2/s^{-1}			
Concrete Sample	Data No.	Middle Value	Standard Deviation
T-1	5	$1.78x10^{-12}$	$0.30x10^{-12}$
T-2	5	$1.45x10^{-12}$	$0.23x10^{-12}$

Table 2 MCI® Coefficient of Diffusion

		Product	Description	Approximate Dosage Rate	Packaging
Admixtures	Amino-Alcohol Based	MCI®-2000	Liquid, aminoalcohol based concrete admixture. Patented.	1 pt/yd^3 (0.62 l/m^3)	5 gal (19 l) pails, 55 gal (208 l) drums
		MCI®-2001	Powder, fumed silica/MCI®-2000 combination. Patented.	3 lb/yd^3 (1.78 kg/m^3)	5 lb (2.3 kg) boxes, 50 lb (22.7 kg) and 100 lb (45.4 kg) drums
	Amine Carboxylate Based	MCI®-2005	Liquid, amine carboxylate based concrete admixture. Can retard concrete setting time 3-4 hours at 70ºF (21ºC). Patented.	1 pt/yd^3 (0.6 l/m^3)	5 gal (19 l) pails, 55 gal (208 l) drums 275 gal (1040 l) totes
		MCI®-2005 NS	Liquid, normal set version of MCI®-2005. Can not be frozen. Patented.	1.5 pts/yd^3 (1 l/m^3)	5 gal (19 l) pails, 55 gal (208 l) drums 275 gal (1040 l) totes
		MCI®-2005 AL	Liquid, normal set version of MCI®-2005 with less ammonia odor. Patented.	1.5 pts/yd^3 (1 l/m^3)	5 gal (19 l) pails, 55 gal (208 l) drums 275 gal (1040 l) totes
		MCI®-2006	Powder, amine carboxylate based concrete admixture. Can retard concrete setting time 3-4 hours at 70ºF (21ºC). Patented.	1 lb/yd^3 (0.6 kg/m^3)	5 lb (2.3 kg) boxes, 50 lb (22.7 kg) and 100 lb (45.4 kg) drums.
		MCI®-2006 NS	Powder, normal set version of MCI®-2006. Patented.	1 lb/yd^3 (0.6 kg/m^3)	5 lb (2.3 kg) boxes, 50 lb (22.7 kg) and 100 lb (45.4 kg) drums.
	Specialty	MCI®-2012 IntegRepel	A concrete waterproofing admixture designed for protection of steel reinforcement .	1.7 qt/yd^3 (2.1 l/m^3)	5 gal (19 l) pails, 55 gal (208 l) drums, 275 gal (1040 l) totes
		MCI® Grenades®	MCI®-2006 NS powder pre-measured into water soluble bags for admixing into concrete.	1 grenade/ yd^3	20 grenades/ carton
		MCI® Grenades®-Metric	MCI®-2006 NS powder pre-measured into water soluble bags for admixing into concrete.	1 grenade/ m^3	20 grenades/ carton
		MCI® Mini Grenades®	MCI®-2006 NS powder pre-measured into water soluble bags for admixing into concrete.	1 per 0.5-0.6 ft^3 (1 per 0.0015 m^3)	100 grenades/ carton
		MCI® Fiber Grenades®	MCI®-2006 NS powder and MCI® Fibers pre-measured into water soluble bags for admixing into concrete.	2 grenades/ yd^3	20 grenades/ carton
		MCI® Metric Fiber Grenades®	MCI®-2006 NS powder and MCI® Fibers pre-measured into water soluble bags for admixing into concrete.	2 grenades/ m^3	20 grenades/ carton
		MCI® Fibers	Mono-filament polypropylene-based fibers containing Migratory Corrosion Inhibitors (MCI®).	1.5 lbs/yd^3 (910 g per m^3)	5 lb (2.3 kg) boxes, 50 lbs (22.7 kg) and 100 lbs (45.4 kg) drums
	Superplasticizers with Amine Carboxylate Based MCI®	MCI®-2007 SuperCorr™	Liquid, melamine based superplasticizer with MCI®. Patented.	3-4 pts/yd^3 (1.5-2 l/m^3)	5 gal (19 l) pails, 55 gal (208 l) drums
		MCI®-2007 P	Powder, polycarboxylate based superplasticizer with MCI®, which reduces mixing water by 20-30% depending on cement type.	16-27 oz/yd^3 (0.6 - 1.0 kg/m^3)	5 lb (2.3 kg) boxes, 50 lbs (22.7 kg) and 100 lbs (45.4 kg) drums
		MCI®-2008 ViaCorr™	Powder, polycarboxylate based superplasticizer for self compacting, self leveling concrete with MCI®.	0.4-0.6% by weight of concrete mix	50 lbs (22.7 Kg) and 100 lbs (45.4 Kg) drums
		MCI®-2008 L	Liquid, polycarboxylate based superplasticizer for self compacting, self leveling concrete with MCI®.	0.4-0.6% by weight of concrete mix	5 gal (19 l) pails 55 gal (208 l) drums

		Product	Description	Approximate Dosage Rate	Packaging
Surface Treatments	Amine Carboxylate Based	MCI®-2020/ MCI®-2020 V/O	Clear, penetrating surface treatment for existing structures. Contains Migrating Corrosion Inhibitors that form a protective film on embedded metals. UL approval to meet NSF Standard 61 Certification for use on structures holding potable water. V/O Version for vertical and overhead applications.	150 ft^2/gal (3.68 m^2/l)	5 gal (19 l) pails 55 gal (208 l) drums
		MCI®-2020 M/ MCI®-2020 M V/O	New, ready to use version of MCI®-2020 that provides even better corrosion protection. V/O Version for vertical and overhead applications.	150 ft^2/gal (3.68 m^2/l)	5 gal (19 l) pails 55 gal (208 l) drums
		MCI®-2020 M SC	New, concentrated version of MCI®-2020 that provides even better corrosion protection. Dilute 1:1 with water to make ready to use product.	150 ft^2/gal (3.68 m^2/l)	5 gal (19 l) pails 55 gal (208 l) drums
		MCI®-2020 Powder/ MCI®-2020 V/O Powder	Powder version of MCI®-2020. One 100 lb (45.35 kg) drum makes 55 gallons (208 liters) of MCI®-2020 liquid. V/O Version for vertical and overhead applications.	150 ft^2/gal (3.68 m^2/l)	100 lb (45.4 kg) drums
	Water Repellants with MCI®	MCI®-2018/ MCI®2018 V/O	100% solids, organosilane sealer containing MCI®. Spray, brush or roller applied.	125-175 ft^2/gal (3 -4.2 m^2/l)	5 gal (19 l) pails 55 gal (208 l) drums
		MCI®-2019	VOC-exempt solvent based 40% silane sealer containing MCI®. Spray, brush or roller applied.	125-175 ft^2/gal (3 - 4.2 m^2/l)	5 gal (19 l) pails 55 gal (208 l) drums
		MCI®-2019 W	Water based, 40% silane sealer containing MCI®. Spray, brush or roller applied.	125-175 ft^2/gal (3 - 4.2 m^2/l)	5 gal (19 l) pails 55 gal (208 l) drums
		MCI®-2021	Water based, silicate sealer containing MCI®. Spray, brush or roller applied. Patented.	150-250 ft^2/gal (3.7-6.1 m^2/l)	5 gal (19 l) pails 55 gal (208 l) drums
		MCI®-2022/MCI®-2022 V/O	Water based, silane/siloxane blend sealer containing MCI®. Spray, brush or roller applied. V/O Version for vertical and overhead applications. Patented.	125-175 ft^2/gal (3-4.2 m^2/l)	5 gal (19 l) pails 55 gal (208 l) drums
		MCI®-POWR 40	Silane based, penetrating, oil and water repellant containing MCI®. Spray, brush or roller applied.	125-175 ft^2/gal (3 - 4.2 m^2/l)	5 gal (19 l) pails 55 gal (208 l) drums
		MCI®-POWR 100	Silane based, penetrating, oil and water repellant containing MCI®. Spray, brush or roller applied.	125-175 ft^2/gal (3 - 4.2 m^2/l)	5 gal (19 l) pails 55 gal (208 l) drums
	Coatings	MCI®- 2026 Primer HS	Two-component, chemically resistant, water based, epoxy primer for concrete.	250-350 ft^2/gal (6.1-8.5 m^2/l)	0.75 gal (2.3 l), 6 gal (22.7 l), 15 gal (56.8 l),165 gal (624.6 l) yield kits.
		MCI®-2026 Floor Coating	Two-component, chemically resistant, 100% solids Novolac epoxy for concrete with excellent chemical and abrasion resistance.	125-150 ft^2/gal (3.0-3.7 m^2/l)	0.6 gal (2.27 l), 5 gal (19 l), 12.5 gal (47.3 l), 138 gal (522.4 l) yield kits.
		MCI®-2027 Polyurea	Single-component polyurea that is fast drying, UV stable, and has a high gloss finish. Available in clear, grey, tan, and white.	2-3 coats of 10 mils DFT = 112-125 ft^2/gal (2.7-3 m^2/l)	5 gal (19 l) pails 55 gal (208 l) drums.
		MCI®-2241/2242	Flexible and breathable waterproofing membranes based on a unique combination of acrylic emulsion, Portland cement and fine fibers. MCI®-2241 is a grey color, MCI®-2242 is white.	Each kit covers 88-100 ft^2 when applied at 1/16 inch thick (8 m^2 per 15.2 l at 1.6 mm)	Each kit yields 4 gallons (15 l). Component A is 2.3 gal (8.9 l) packaged in 5 gal (19 l) pail. Component B is 25 lb (11 kg) bags.
		MCI®-EcoRainbow Architectural Coating	Clear, water-based, acrylic primer/top coat containing MCI®. Also available in white, grey, and custom colors.	535-641 ft^2/gal (13-16 m^2/l)	5 gal (19 l) pails 55 gal (208 l) drums.
		MCI®-Anti-Graffiti Coating	Two-component, solvent based aliphatic urethane for concrete to provide easy removal of graffiti.	516 ft^2/gal at 2 mils DFT (13 m^2/l at 50 microns DFT)	Two 5 gal (19 l) pails (Part A and Part B) per kit.
		MCI®-Wall Defense	Clear, silicone elastomer based, anti-graffiti coating for concrete, masonry, and other metal surfaces. MCI® Wall Defense does not need to be reapplied after graffiti removal.	80-100 ft^2 at 12-15 mils DFT (2.0-2.5 m^2/l at 300-375 microns DFT)	5 gal (19 l) pails 55 gal (208 l) drums.

	Product	Description	Approximate Dosage Rate	Packaging
Specialty Products	VpCI®-422/VpCI®-423/VpCI®-426	Water-based rust removers. Removes rust stains from concrete. Also available in gel form. Rinse concrete with MCI®-2060 after application to neutralize.	200-600 ft^2/gal (5-15 m^2/l)	5 gal (19 l) pails, 55 gal (208 l) drums, liquid totes and bulk.
	MCI® Super Remover	Water-based cleaner for removal of calcium carbonate and oxides, including hardened concrete, rust on mild steels, and encrusted hard water scales.	May be used as is or diluted up to 25%	5 gal (19 l) pails, 55 gal (208 l) drums, liquid totes and bulk.
	MCI® CorShield	Water-based coating or embedded reinforcement as well as offering long term (5+ years) indoor storage or short/medium term (6-24 months) outdoor storage protection.	187-561 ft^2/gal (4.6-14 m^2/l)	
	VpCI®-432/433	Paint stripper and graffiti remover. Removes paint from concrete without damage. Non-caustic, non-toxic, water cleanable. Also available in gel form.	200-800 ft^2/gal (5-20m^2/l)	5 gal (19 l) pails, 55 gal (208 l) drums, liquid totes and bulk.
	MCI®-2060	Cleaner and degreaser that contains MCI®. It effectively cleans caked on grease, dirt, oil and mud off of concrete.	May be used as is or diluted up to 1%	5 gal (19 l) pails, 55 gal (208 l) drums.
	MCI®-2061	Cleaner and degreaser containing MCI®. Contains microorganisms that break down oils and other petroleum based materials.	Use as concentrate on oil stains, 24-48 oz/gal water for cleaning (188-375 ml/l)	5 gal (19 l) pails, 55 gal (208 l) drums.
	GalvaCorr®	Four component, galvanic protection coating for concrete structures. Based on technology developed at NASA Kennedy Space Center. Installed on non-traffic bearing surfaces.	60 ft^2/gal @ 24 mils (1.6 m^2/l @ 600 µM)	50 lb/5 gal (22.7 kg/19 L) kits
	VpCI® CorrVerter®	Water-based primer for rusted or poorly prepared surfaces. Does NOT contain tannic or phosphoric acid.	167-278 ft^2/gal (4.2-5.6 m^2/l)	5 gal (19 l) pails.
	MCI® Coating for Rebar	Water based, barrier coating that provides extended outdoor protection for exposed steel and aluminum.	300 ft^2/gal (7.3 m^2/l)	5 gal (19 l) pails 55 gal (208 l) drums.
	MCI® Coating for Rebar NT	Non-tacky version of MCI® Coating for Rebar.	300 ft^2/gal (7.3 m^2/l)	5 gal (19 l) pails 55 gal (208 l) drums.
	MCI® Peel Off Coating	Temporary, removable coating containing MCI® for protection against nicks, abrasion, scratches, etc.	140-160 ft^2/gal @ 4 mils (13-15 m^2/l @ 100 microns)	5 gal (19 l) pails 55 gal (208 l) drums.
	MCI®-2050	Form/mold release agent containing MCI® technology. Forms a thin protective film to which concrete, asphalt, dirt or other debris will not stick.	Rates vary dependent upon substrate & desired results. ~125-150 ft^2/gal (3.0-3.7 m^2/l)	5 gal (19 l) pails, 55 gal (208 l) drums, liquid totes and bulk.
	MCI® Creteskin™	An industrial strength release agent containing MCI®. This clear protective coating inhibits the adhesion of concrete and other materials on painted and unpainted metal surfaces.	320-640 ft^2/gal @ 0.5-1 mil DFT, (8-16 m^2/l @ 12.5-25 microns)	5 gal (19 l) pails, 55 gal (208 l) drums, liquid totes and bulk.
	MCI®-309	A corrosion inhibiting powder for protection of ferrous metals in recessed areas, interior cavities and voids.	0.3-0.5 oz/ft^3 (300-500 g/m^3)	5 lb (2.3 kg) boxes, 50 lb (22.7 kg) and 100 lb (45.4 kg) drums.
	PTC Emitters	Tyvek® pouches filled with MCI® powder for corrosion protection of post tensioned cables and other metallic components in recessed areas, interior cavities, and voids.	1 pouch protects 35 ft^3 (1m^3)	Carton of 50 pouches
	MCI®-2005 Gel	MCI®-2005 in gel format for injection into existing structures.	Based on hole size and number of holes	13 oz (384ml) tubes, 5 gal (19 l) pails, 55 gal (208 l) drums.
	MCI®-2020 Gel	MCI®-2020 in gel format for injection into existing structures.	Based on hole size and number of holes	13 oz (384ml) tubes, 5 gal (19 l) pails, 55 gal (208 l) drums.
	MCI®-2070	An admixture for asphalt primers used in concrete highway or street asphalt overlays to increase adhesion and effectively inhibit corrosion occurring within the concrete.	Add at a concentration level of 2-4% by weight	5 gal (19 l) pails, 55 gal (208 l) drums
	MCI®-Construction Film	A polyethylene film designed for use in the construction industry which inhibits corrosion on both ferrous and nonferrous metals.	n/a	20' x 100' sheeting, 4 mil (6.1 m x 30.48 m, 100 microns)

	Product	Description	Approximate Coverage Rate	Packaging
Repair Products	MCI®-2023	MCI® passivating repair grout for protecting reinforcing steel in concrete. Patent Pending.	60 ft^2 at a thickness of $^1/_{16}$ inch (5.6 m^2 at a 1.6 mm thickness)	Part A 11 lb (5kg) resin, Part B 26.5 lb (12 kg) bag powder. Yields 2.5 gal (9.4 l)
	MCI®-2246	MCI® bonding agent is a unique combination of Portland cement, microsilica, epoxy, and acrylic resin.	70-80 ft^2/gal at 20 mils WFT (1.6-1.8 m^2/l at 0.5 mm)	2 part kit includes 1 gal jug (3.8 l) and 28 lb (13 kg) bag
	MCI®-Mini Grenades	MCI®-2006 NS powder pre-measured into water soluble, PVA bags. Allows you to add corrosion inhibitor to any bagged mortar or grout mix.	1 per 0.5-0.6 ft^3 (1 per 0.0015 m^3)	100 grenades/ carton
	MCI®-2039 SC	Single component, fiber reinforced MCI® repair mortar containing MCI®-Mini Grenades, fibers and polymers.	12 ft^2 per bag at a ½ inch thick (1.12 m^2 at 12.7 mm thick).	55 lb (25kg) bag of powder
	MCI®-2701	Single component, trowel grade repair mortar with MCI®-2006 NS technology. It is a polymer-modified cement-based mortar for structurally repairing or overlaying deteriorated concrete.	20-25 ft^2 per bag at ¼ inch thick (2 m^2 at 6 mm thick).	60 lb bag yields 0.5 ft^3 (27 kg bag yields 0.014 m^3).
	MCI®-2702	Single component, overhead repair mortar with MCI®-2006 NS technology.	20-25 ft^2 per bag at ¼ inch thick (2 m^2 at 6 mm thick).	50 lb bag yields 0.45 ft^3 (23 kg bag yields 0.013 m^3)

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